

AI SMART SHOP
AI-POWERED SMART RETAIL & INVENTORY
MANAGEMENT SYSTEM

PROJECT REVIEW – 1

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Project Type: Software / AI-Based Application
Track: Continuation Track

Project Details

Detail	Information
Project Title	AI SMART SHOP – AI-Powered Smart Retail & Inventory Management System
Project Type	Software / AI-Based Application
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Department	Computer Science and Engineering
Institution	Rathinam Technical Campus, Coimbatore
Academic Year	2026–2027
Review	Project Review – 1
Development Track	Continuation Track

Problem Statement

"The existing retail stock system automatically updates inventory when products are received or sold, but it does not proactively alert the shopkeeper when stock is approaching a critical level, while access to important digital operations may depend on a worker who is comfortable with computer-based interfaces and English typing, causing difficulty and potential delays when the worker is unavailable."

Retail shops need to manage many products, changing stock levels, sales, purchases and suppliers every day. When stock information is handled manually or across disconnected records, it can be difficult to quickly identify which products are running low, which items need immediate replenishment and how current transactions are affecting stock.

A second practical challenge is accessibility. Some shopkeepers and workers may not be comfortable with English typing or with operating complex computer-based systems. A useful digital solution should therefore be simple to understand, visually clear and capable of more natural interaction.

The project addresses these challenges by bringing retail operations into one interface and by connecting stock awareness with actionable alerts and replenishment workflows.

Proposed AI-Based Solution

AI SMART SHOP proposes an AI-assisted retail management system that combines inventory management, stock intelligence, sales, purchases, supplier management, alerts, smart reorder, analytics and an AI assistant.

The proposed approach is designed to help the user understand what is happening in the shop and what action may be needed next, rather than simply storing transaction records.

- Monitor product quantities against configured stock thresholds.
- Classify stock conditions as Normal, Low or Critical.
- Highlight Low and Critical items through alerts.
- Connect stock alerts to a Smart Reorder / replenishment workflow.
- Update inventory automatically when sales or purchases are recorded.
- Use analytics to present useful business and stock information.
- Provide AI-assisted natural-language interaction for inventory-related queries.

How AI SMART SHOP Works

The current prototype follows a connected retail workflow. Each major operation affects the relevant inventory information, allowing the user to move from identifying a problem to taking an action.

Stage	How It Works
1. Product Management	Products are added to the inventory with relevant stock information and thresholds.
2. Stock Intelligence	The system evaluates the available quantity and identifies Normal, Low or Critical stock.
3. Alerts	Low and Critical conditions are surfaced so that the user can notice products requiring attention.
4. Smart Reorder	The alert provides a direct path toward a reorder / purchase action.
5. Purchase Update	A purchase increases stock and the stock condition is recalculated.
6. Sales Update	A sale is validated against available stock and the inventory quantity is reduced.
7. Analytics	Sales and purchase information is used to present business and inventory insights.
8. AI Interaction	The AI assistant accepts inventory-related natural-language input; voice input can be converted to text and responses can be spoken through text-to-speech.

What Has Been Completed So Far

The project was first developed with a basic website/application structure so that each major retail operation has a clear place in the system. This basic structure has now been extended into a functional prototype with connected workflows.

- Basic responsive application structure with separate areas for Dashboard, Inventory, Sales, Purchases, Suppliers, Alerts, Analytics and AI Assistant.
- Dashboard structure for viewing stock and business information, alerts and quick actions.
- Inventory structure with product management, search, filtering, stock thresholds and stock adjustment.
- Sales workflow for recording transactions and updating stock.
- Purchase workflow for recording purchases and increasing stock.
- Supplier management functionality.
- Dynamic stock intelligence using Normal, Low Stock and Critical Stock conditions.
- Low-stock and critical-stock alert functionality.
- Smart Reorder / Reorder Now action connected to stock alerts.
- Business and inventory analytics.
- AI Smart Assistant for inventory-related natural-language interaction.
- Voice recognition for spoken input and text-to-speech for audio responses.
- Responsive interface suitable for desktop and mobile screens.

Key Features and Modules Completed

Module / Feature	Main Functionality	Status
Dashboard	Stock overview, business indicators, alerts and quick actions	Completed
Inventory Management	Add, edit, delete, search, filter, thresholds and stock adjustment	Completed
Sales Management	Sale entry, stock validation, total calculation and stock deduction	Completed

Purchase Management	Purchase entry, supplier selection, cost and stock increase	Completed
Supplier Management	Supplier information and management	Completed
Stock Intelligence	Normal / Low / Critical stock classification	Completed
Alerts & Notifications	Low-stock and critical-stock alerts	Completed
Smart Reorder	Direct reorder action from stock alerts	Completed
Analytics	Revenue, purchase cost, estimated profit, valuation, movement and stock health	Completed
AI Smart Assistant	Natural-language inventory-related interaction	Working
Voice Interaction	Speech-to-text input and text-to-speech response	Working
Responsive UI	Desktop and mobile-friendly interface	Completed

What Is Currently Working

The following functions are currently demonstrated in the prototype:

- Products can be added, edited, deleted, searched and filtered.
- Stock can be adjusted and its condition changes according to configured thresholds.
- The system identifies Normal, Low Stock and Critical Stock conditions.
- Low and Critical stock conditions generate visible alerts.
- Users can initiate Smart Reorder from the stock-alert workflow.
- Sales validate stock availability and reduce inventory after a successful transaction.
- Purchases increase inventory and update stock condition.
- Supplier information can be managed.
- Analytics displays business and inventory-related information.
- The AI Smart Assistant accepts inventory-related natural-language queries.
- Voice recognition converts spoken input into text.
- Text-to-speech provides audio responses.
- The interface adapts to desktop and mobile screen sizes.

How the Project Is Useful to People

- Helps shopkeepers understand stock condition without repeatedly checking every product manually.
- Makes products requiring attention easier to identify through Low and Critical alerts.
- Connects an identified stock problem with a practical reorder action.
- Reduces the gap between sales/purchases and inventory updates by connecting these workflows.
- Provides a single interface for products, suppliers, transactions and business insights.
- Makes important information easier to understand through dashboards, status indicators and analytics.
- AI-assisted interaction can make it easier to ask inventory-related questions in natural language.
- Future multilingual and expanded voice interaction can make the system more accessible to people who are less comfortable with English typing or conventional computer interaction.
- Enable shopkeepers to access stock, sales, billing and purchase information through multilingual voice commands without relying on English typing.
- Reduce dependency on computer-skilled workers by providing a simple, mobile-first AI interface for quick shop operations.

Pending Work and Next Steps

The following items are intentionally classified as future work. They are not claimed as completed in this Review 1 report.

Future Area	Planned Development
1. Multilingual Interaction	Add multiple supported languages so shopkeepers and workers who are not comfortable with English typing can interact with the system more easily.
2. Expanded Voice-Based Interaction	Extend voice interaction from basic speech input/output toward practical spoken commands for common shop operations, reducing dependence on

	keyboard and mouse use. Develop multilingual support and expand voice-based interaction.
3. Database Integration	Replace prototype/local data handling with a reliable database so information is persistent, secure and scalable. Implement database integration and improve data persistence.
4. Advanced AI Intelligence	Introduce demand forecasting, sales-pattern analysis, stock prediction and stronger product recommendations.
5. Smarter Reordering	Develop AI-assisted reorder quantity and timing recommendations based on stock and transaction patterns.
6. Barcode / QR Integration	Add faster product identification and inventory operations using barcode or QR scanning.
7. Real-User Validation	Test the prototype with at least three real/new users, collect genuine feedback and document improvements based on that feedback.
8. Multi-User and Role-Based Access	Support different users such as shop owner, staff and administrator with suitable permissions.
9. Security and Production Readiness	Strengthen authentication, validation, data protection, error handling and deployment practices.
10. Reports and Automation	Expand reports, notifications and automated retail-management workflows.
11. Continuous UI/UX Improvement	Refine the interface based on user feedback and make important operations even easier for first-time users.

Completion Position

AI SMART SHOP has reached a **functional prototype stage**, with the core system structure and major inventory management workflows successfully implemented. The

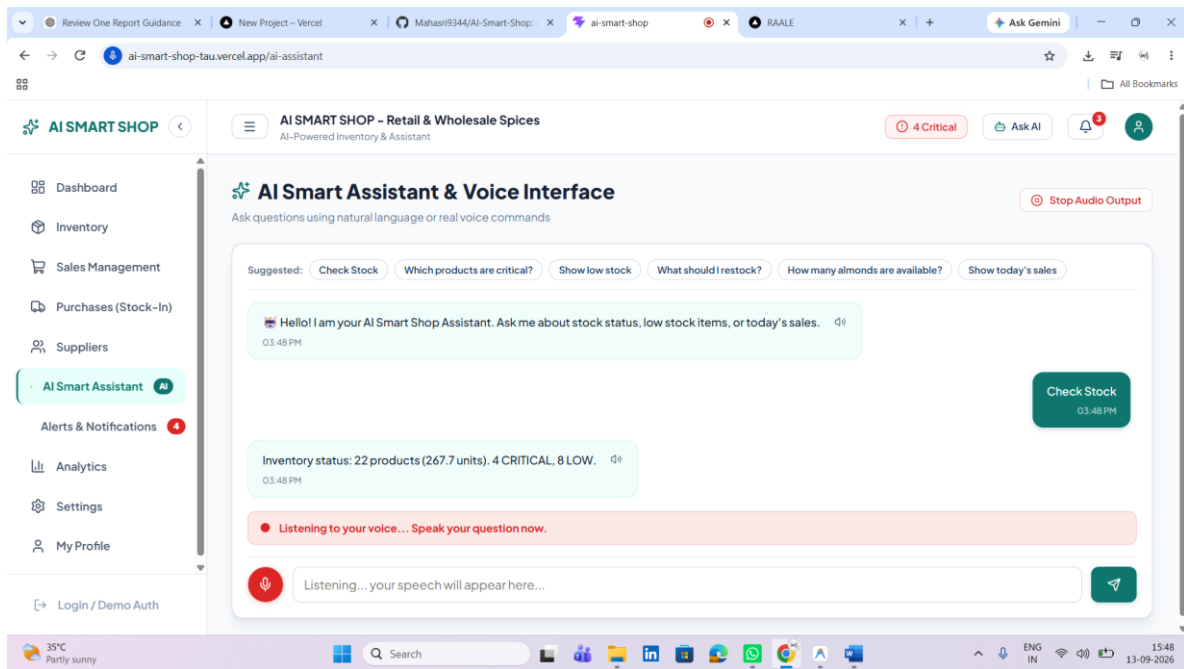
current version demonstrates working modules for inventory, sales, purchases, suppliers, alerts, analytics, and AI-assisted interaction. Further development will focus on multilingual support, enhanced AI capabilities, database integration, user validation, accessibility improvements, and production-level deployment.

Review 1 Requirement	AI SMART SHOP Position
What has been completed	Basic website/application structure and major core retail workflows implemented.
Key features/modules	Dashboard, Inventory, Sales, Purchases, Suppliers, Stock Intelligence, Alerts, Smart Reorder, Analytics, AI Assistant and Voice Interaction.
What is currently working	Core inventory, transactions, stock status, alerts, reorder, analytics, AI interaction, voice input/audio response and responsive interface.
Pending work / next steps	Multilingual support, expanded voice commands, database, advanced AI, smarter reorder, barcode/QR, real-user validation, multi-user access, security and production improvements.

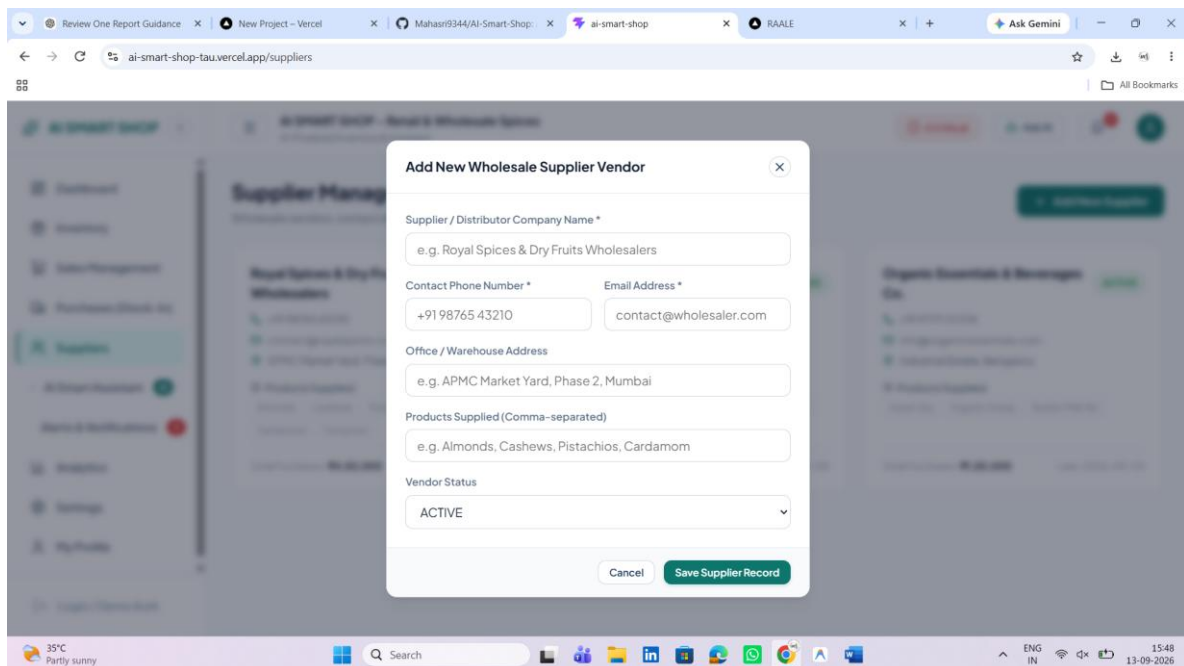
Project Repository & Live Demonstration

<https://github.com/Mahasri9344/AI-Smart-Shop>

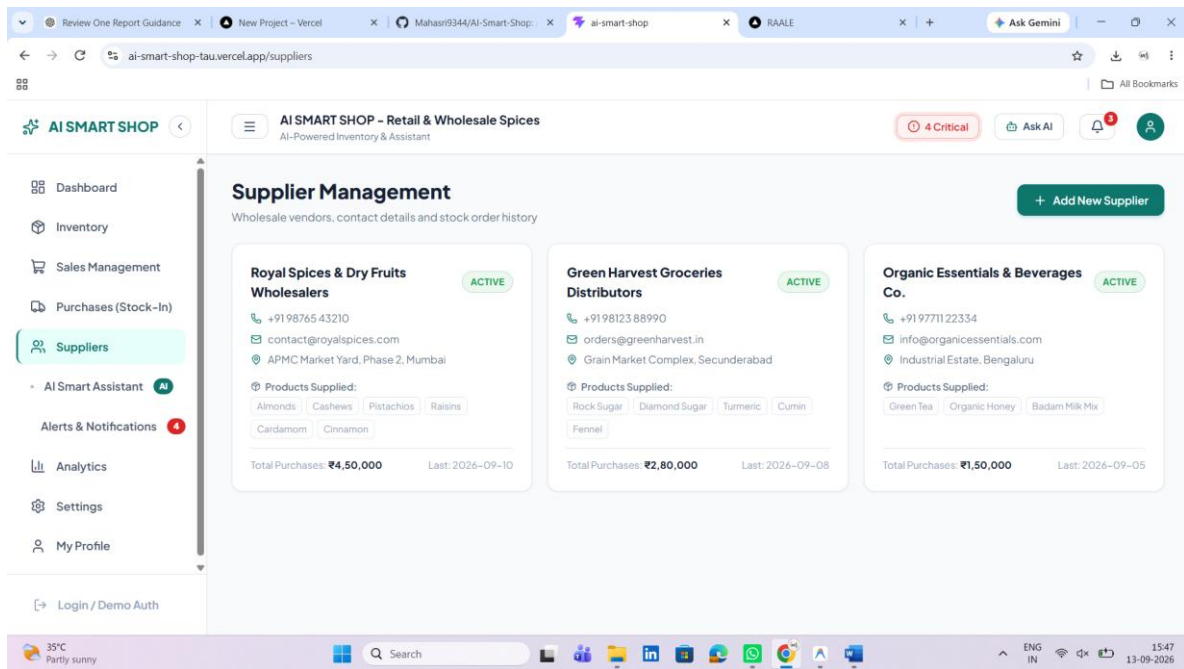
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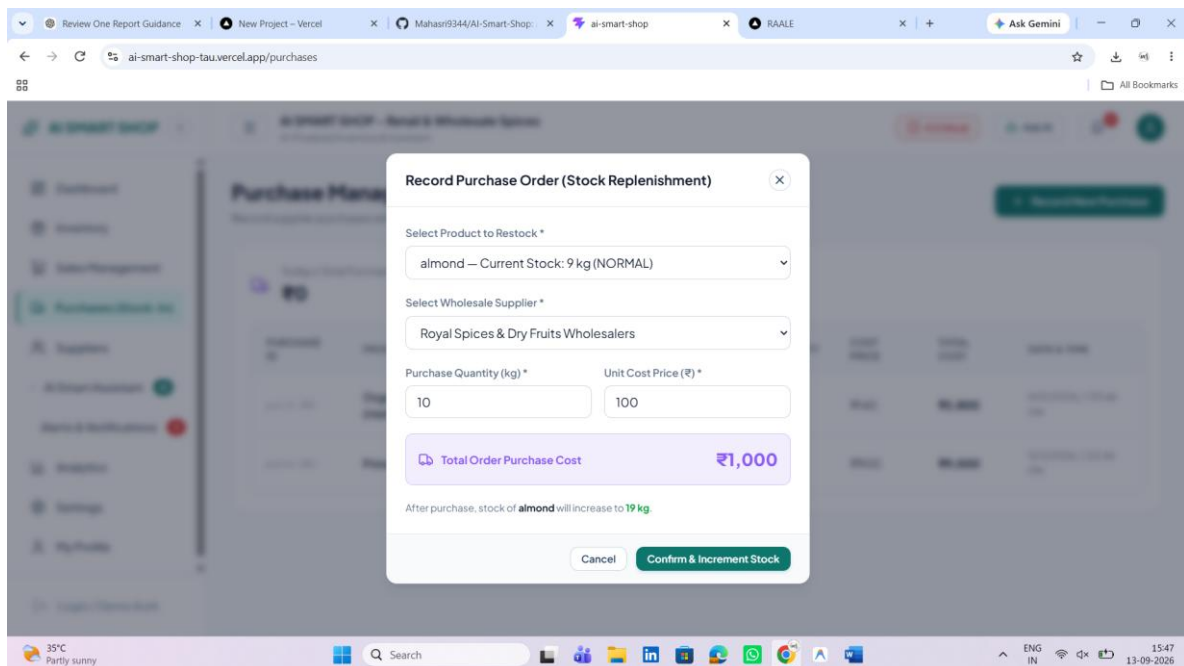
AI Smart Assistant & Voice Interface – Natural-language inventory interaction with voice input and audio response.



Add New Wholesale Supplier Vendor – Supplier registration form with contact, warehouse and product details.



Supplier Management – Wholesale supplier records, supplied products and purchase history.



Purchase Replenishment Workflow – Product, supplier, purchase quantity, cost and updated stock preview.

AI SMART SHOP - Retail & Wholesale Spices
AI-Powered Inventory & Assistant

Purchase Management (Stock-In)
Record supplier purchases with automatic inventory replenishment

Today's Total Purchases: ₹0

PURCHASE ID	PRODUCT	SUPPLIER	QUANTITY	COST PRICE	TOTAL COST	DATE & TIME
purch-201	Organic Turmeric Powder (Haldi)	Green Harvest Groceries Distributors	20 kg	₹140	₹2,800	9/12/2026, 1:33:46 PM
purch-202	Pistachios (Pista Salted)	Royal Spices & Dry Fruits Wholesalers	10 kg	₹900	₹9,000	9/11/2026, 1:33:46 PM

Purchase Management (Stock-In) – Recorded supplier purchases with quantity, cost and automatic stock replenishment.

Record Customer Sale

Select Product *

almond — Current Stock: 9 kg (₹150/kg)

Product Info & Stock Available Quantity

almond (Dry Fruits) 9 kg

Sale Quantity (kg) * Unit Price (₹)

1 150

Calculated Total Amount ₹150

Cancel Confirm & Save Sale

Record Customer Sale – Sales entry with stock availability, quantity, unit price and calculated total amount.

AI SMART SHOP – Retail & Wholesale Spices
AI-Powered Inventory & Assistant

Sales Management
Record customer sales with automatic stock deduction & revenue tracking

Today's Total Sales: ₹5,050

TRANSACTION ID	PRODUCT	QUANTITY	UNIT PRICE	TOTAL AMOUNT	DATE & TIME
sale-178928685913	almond	1 kg	₹150	₹150	9/13/2026, 1:34:45 PM
sale-181	Cashews (Kaju W240)	2 kg	₹950	₹1,900	9/13/2026, 11:33:46 AM
sale-182	Almonds (Badam)	1 kg	₹900	₹900	9/13/2026, 9:33:46 AM
sale-183	Rock Sugar (Dhaga Mishri)	5 kg	₹110	₹550	9/13/2026, 7:33:46 AM
sale-184	Green Cardamom (Elaichi 8mm)	0.5 kg	₹3,100	₹1,550	9/13/2026, 5:33:46 AM

Sales Management – Customer sales transactions with quantities, prices, total amounts and date/time.

Add New Product

Product Name *
e.g. Kashmiri Almonds (Badam)

Category: Dry Fruits Unit Type: Kilograms (kg)

Description / Grade
e.g. Whole King-size Kernels

Stock Quantity: 10 Cost Price (₹): 100 Selling Price (₹): 150

Low-Stock Threshold: 8 Critical Threshold: 3

Supplier / Distributor
Royal Spices & Dry Fruits Wholesalers

Buttons: Cancel, Add Product

Add New Product – Product registration with category, unit type, stock quantity, prices and stock thresholds.

AI SMART SHOP - Retail & Wholesale Spices
AI-Powered Inventory & Assistant

Inventory Management
Real-time stock tracking with dynamic status computation

Search products by name, category, or supplier...

PRODUCT NAME	CATEGORY	QUANTITY	THRESHOLDS (LOW / CRITICAL)	PURCHASE PRICE	SELLING PRICE	STATUS	ACTIONS
almond whole king	Dry Fruits	9 kg	Low: 8 kg Crit: 3 kg	₹100	₹150	NORMAL	[Edit] [Delete]
Almonds (Badam) Premium California Premium Almonds	Dry Fruits	2 kg	Low: 10 kg Crit: 3 kg	₹650	₹900	CRITICAL	[Edit] [Delete]
Cashews (Kaju W240) Whole King-size Cashews	Dry Fruits	15 kg	Low: 12 kg Crit: 4 kg	₹700	₹950	NORMAL	[Edit] [Delete]
Green Raisins (Kishmish) Long Sweet Green Raisins	Dry Fruits	5 kg	Low: 8 kg Crit: 3 kg	₹220	₹320	LOW	[Edit] [Delete]
Walnuts (Akhrot Giri) Kashmiri Soft Kernel Walnuts	Dry Fruits	1.5 kg	Low: 6 kg Crit: 2 kg	₹850	₹1,200	CRITICAL	[Edit] [Delete]

Inventory Management – Product search, filtering, stock quantities, thresholds and Normal/Low/Critical status.

AI SMART SHOP - Retail & Wholesale Spices
AI-Powered Inventory & Assistant

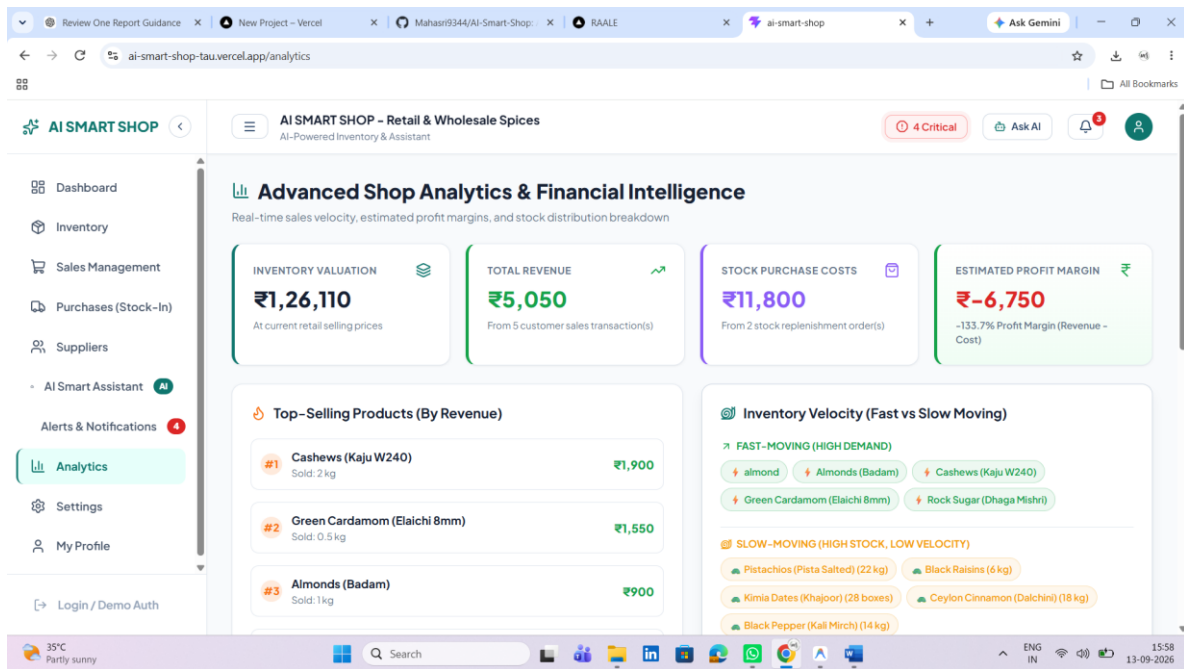
Smart Shopkeeper Dashboard
Real-time stock status, proactive critical alerts & intelligent assistant

PROACTIVE ALERT: 4 CRITICAL STOCK ITEM(S) DETECTED
Almonds (Badam) (2 kg left, critical threshold: 3 kg) • Walnuts (Akhrot Giri) (1.5 kg left, critical threshold: 2 kg) • Dried Figs (Anjeer) (0.8 kg left, critical threshold: 1.5 kg) • Poppy Seeds (Khas Khas) (1.2 kg left, critical threshold: 1.5 kg)

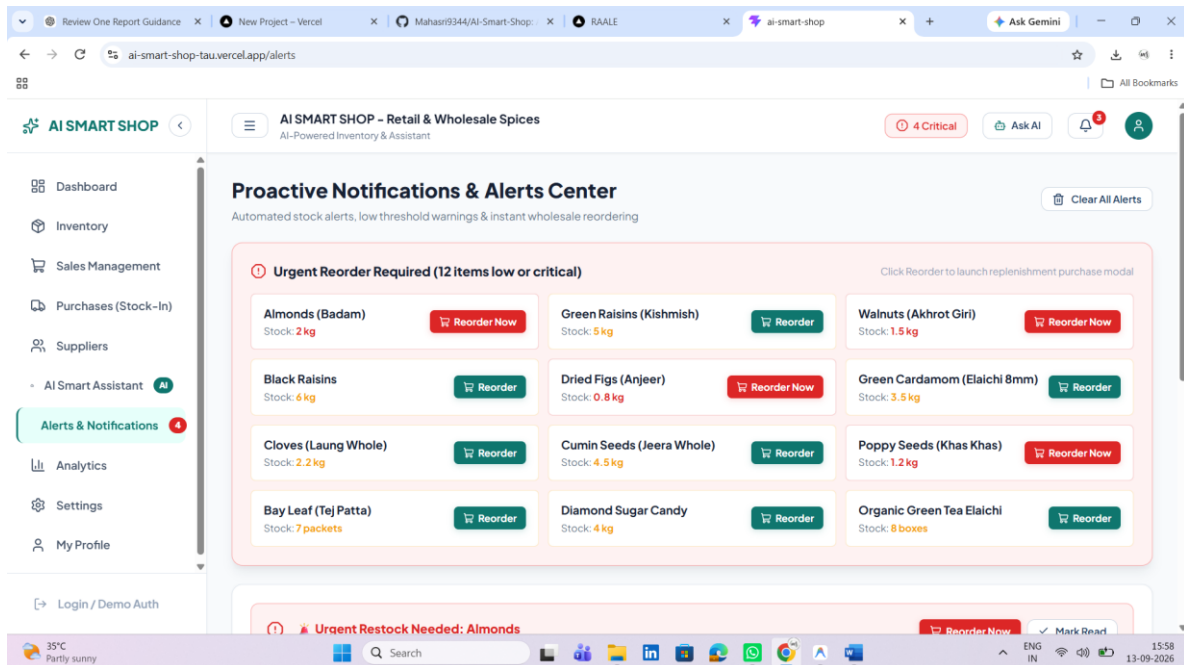
QUICK ACTIONS
+ Add Product | Record Sale | Record Purchase | Ask AI Smart Assistant

Indicator	Value	Description
TOTAL PRODUCTS	22	Across 6 Categories
TOTAL STOCK UNITS	267.7 units	Live Physical Quantity
INVENTORY VALUATION	₹1,26,110	At Current Selling Price
LOW STOCK	8	Approaching Reorder Level
CRITICAL STOCK	4	Immediate Action Required

Smart Shopkeeper Dashboard – Stock overview, proactive critical alerts, key indicators and quick actions.



Advanced Shop Analytics & Financial Intelligence – Revenue, purchase costs, inventory valuation, profit margin and product movement.



Proactive Notifications & Alerts Center – Low/critical stock alerts with direct reorder actions.

Conclusion

AI SMART SHOP has progressed from a basic website structure to a functional AI-assisted retail and inventory management prototype. The current system provides a connected workflow for product management, stock intelligence, sales, purchases, suppliers, alerts, Smart Reorder, analytics and AI-assisted interaction.

The main value of the project is its action-oriented approach: it does not only show that stock is low or critical; it connects the stock condition to an alert and a possible replenishment action. This can help shopkeepers understand their inventory more quickly and manage routine operations more efficiently.

The next stage will focus on making the system more accessible and intelligent through multilingual support, expanded voice-based interaction, database integration, advanced AI recommendations and forecasting, real-user validation, barcode/QR support, secure multi-user operation and production-level improvements.